

DR ESLAM AHMED
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Trigonometric Functions

Three graphs, four moves, one symmetry. Every question in this topic is read off a sketch — draw it first.

THE TOPIC ON ONE MAP

Trigonometric Functions

01 — The three graphs

$\sin$, $\cos$, $\tan$ period asymptote

02 — Transformations

$a \sin x$ $\sin bx$ $\sin(x + c)$ $+k$

03 — Solving by symmetry

all values in the range in terms of α

04 — Counting solutions

two graphs a huge interval

05 — Modelling with trig

$30 \cos(24t)$ dips

01 The Three Graphs

Learn the five key points of each graph. Every other trig graph is one of these, moved.

WHEN YOU SEE → YOU DO

“sketch ... showing the coordinates where it meets the axes”

→ zeros, y-intercept, max, min

“state the period”

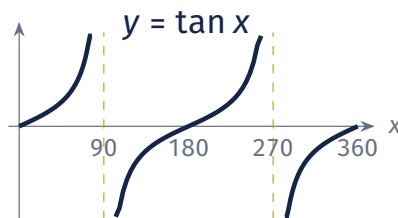
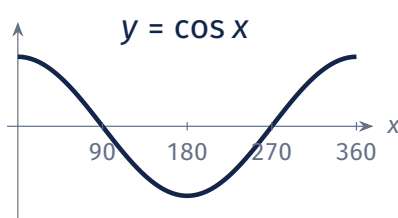
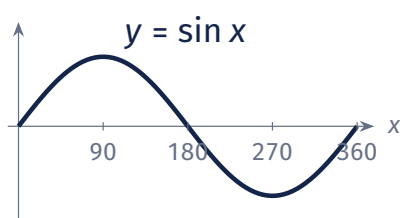
→ sin, cos: 360° (2π) · tan: 180° (π)

“asymptote” to $y = \tan x$

→ $x = 90^\circ + 180^\circ k$ ($\frac{\pi}{2} + k\pi$)

“radians”

→ $\pi = 180^\circ$ — label the axis in multiples of $\frac{\pi}{2}$



x	0	90	180	270	360
sin x	0	1	0	-1	0
cos x	1	0	-1	0	1
tan x	0	–	0	–	0

IDEA 01

Practise: The Three Graphs

My questions first, then selected practice.

Q1 **DR ESLAM** Five key points

P1 • TOPIC 11 3 marks

Sketch $y = \cos x$ for $0 \leq x \leq 360^\circ$, labelling every point where it meets the axes, the maximum and the minimum.

Q2 **DR ESLAM** Periods and asymptotes

P1 • TOPIC 11 2 marks

State the period of $y = \tan x$ and the equations of its asymptotes for $0 \leq x \leq 360^\circ$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 11 6 marks

1 On separate diagrams sketch the graphs of:

(i) $y = \sin x \quad -180^\circ \leq x \leq 180^\circ$

(ii) $y = \cos x \quad 0^\circ \leq x \leq 360^\circ$

(iii) $y = \tan x \quad -180^\circ \leq x \leq 180^\circ$

(6 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 11 3 marks

1 (i) Sketch the graph of $y = \cos \theta$ in the interval $0^\circ \leq \theta \leq 360^\circ$. The sketch must include coordinates of all points where the graph meets the coordinate axes.

(ii) Write down all the values of θ for which $\cos \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

(3 marks)**Q5****PRACTICE**

Practice · Hard

P1 · TOPIC 11 3 marks

3 (i) Sketch the graph of $y = \sin \theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$. The sketch must include coordinates of all points where the graph meets the coordinate axes.

(ii) Given that $\sin 60^\circ = \frac{\sqrt{3}}{2}$, use your graph to find all values of θ in the given interval for which $\sin \theta = -\frac{\sqrt{3}}{2}$.

(3 marks)

02 Transformations: Read a, b, c, k

$y = a \sin(bx + c) + k$ is $y = \sin x$ moved. Each letter does exactly one thing — and you can read it back off a graph.

WHEN YOU SEE → YOU DO

$a \sin x$, “**maximum value of y is ...**”

→ stretch $\parallel y$: $\max = a$, $\min = -a$

$\sin bx$

→ stretch $\parallel x$, s.f. $\frac{1}{b}$; period $\frac{360}{b}$

$\sin(x + c)$

→ translate $\begin{pmatrix} -c \\ 0 \end{pmatrix}$ — inside moves the opposite way

$+k$, “**has a maximum value of 10**”

→ translate up k : new $\max = a + k$

$-\cos x$ / $-\sin x$

→ reflect in the x -axis

“ **P is a minimum point ... find the coordinates of Q** ”

→ count along whole periods

WORKED MODEL • JUNE 2024

$y = 12 \sin x$; $C_2 : y = 12 \sin x + k$ has maximum 3. Find the minimum with the smallest positive x .

1 $12 + k = 3 \Rightarrow k = -9$

outside: y moves

2 min of $\sin x$ first at $x = \frac{3\pi}{2}$

inside unchanged

3 $y = -12 - 9$

$\min = -a + k$

ANSWER $\left(\frac{3\pi}{2}, -21\right)$

IDEA 02

Practise: Transformations

My questions first, then selected practice, then real past papers.

Q6 DR ESLAM Read a and k

P1 • TOPIC 11 2 marks

$y = 4 \cos x + k$ has a maximum value of 7. Find k and the minimum value.

Q7 DR ESLAM Move a point

P1 • TOPIC 11 3 marks

$(90^\circ, 1)$ is the maximum of $y = \sin x$. Write down the maximum point of (a) $y = 3 \sin x$ (b) $y = \sin 2x$ (c) $y = \sin(x - 30^\circ)$

Q8 DR ESLAM Period from b

P1 • TOPIC 11 2 marks

State the period of (a) $y = \cos 3x$ (b) $y = \sin \frac{x}{2}$ (in degrees).

SELECTED PRACTICE

Q9

PRACTICE Practice · Easy

P1 · TOPIC 11 3 marks

- 4 The point P has coordinates $(90^\circ, 1)$ and lies on the graph of $y = \sin x$, where $0^\circ \leq x \leq 180^\circ$.

Write down the coordinates of the image of point P under the following transformations:

(i) $y = f(x) + 2$

(ii) $y = f(3x)$

(iii) $y = f(x + 30^\circ)$

(3 marks)

Q10**PRACTICE**

Practice · Hard

P1 · TOPIC 11**4 marks**

4 (i) Sketch the graph of $y = \tan\left(\theta - \frac{\pi}{4}\right)$ in the interval $-2\pi \leq \theta \leq 2\pi$.

(ii) Write down all the values of θ for which $\tan\left(\theta - \frac{\pi}{4}\right) = 1$ in the given interval.

(4 marks)

Q11**PRACTICE**

Practice · Very Hard

P1 · TOPIC 11 6 marks

10 A function $f(x) = \cos px$, $0 \leq x \leq 2\pi$, first crosses the x -axis at $\frac{\pi}{10}$.

(i) Determine the value of p and sketch the graph of $y = f(x)$.

(ii) State the period of $f(x)$.

(6 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA11 · October 2019 · Q9

P1 · TOPIC 11 4 marks

9.

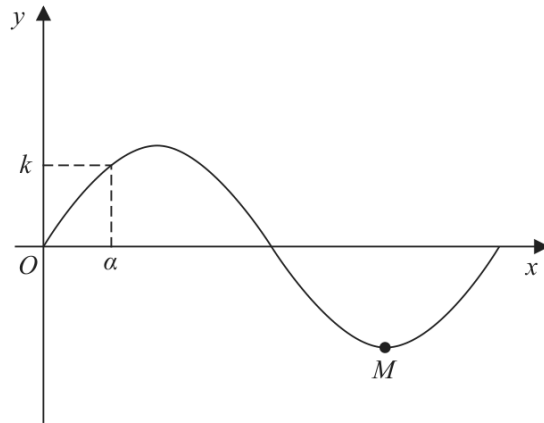


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

- (a) State the period of C . (1)
- (b) State the coordinates of M . (1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

- (c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,
 (ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$. (2)

(Total 4 marks)

Q13**PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q7

P1 · TOPIC 11 5 marks

7.

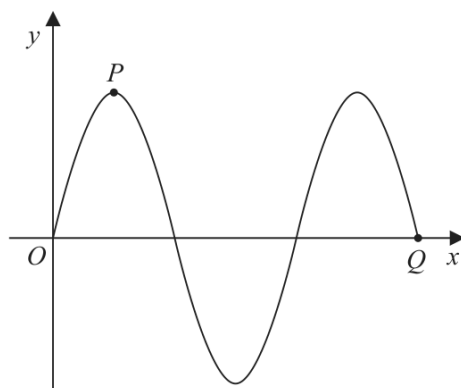
**Figure 3**

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

- (i) the coordinates of P ,
- (ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)**(Total 5 marks)**

Q14 PAST PAPER Pearson IAL WMA11 · January 2021 · Q3

P1 · TOPIC 11 5 marks

3.

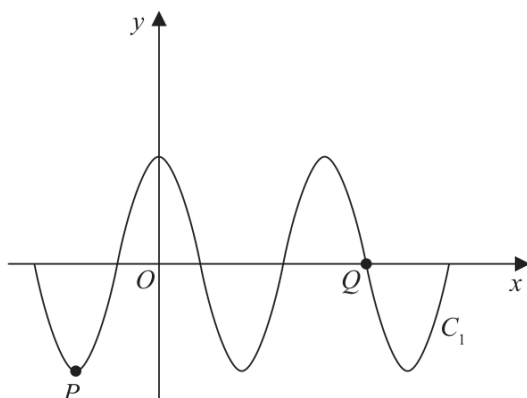
**Figure 1**

Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4\cos x^\circ$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4\cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)**(Total 5 marks)**

Q15**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q4

P1 · TOPIC 11 5 marks

4.

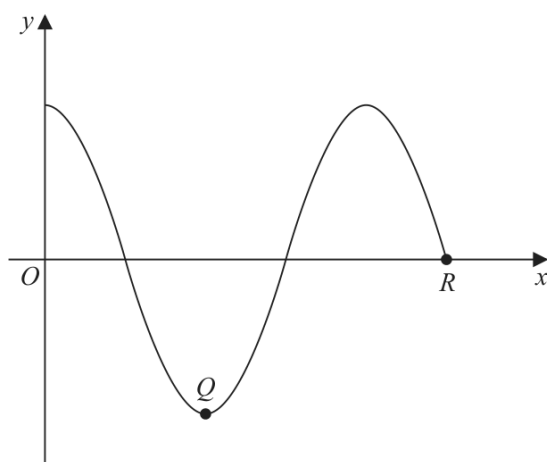
**Figure 2**

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x^\circ \quad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

- (i) the coordinates of Q ,
- (ii) the value of k .

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)**(Total 5 marks)**

Q16**PAST PAPER**

Pearson IAL WMA11 · January 2022 · Q9

P1 · TOPIC 11 4 marks

9.

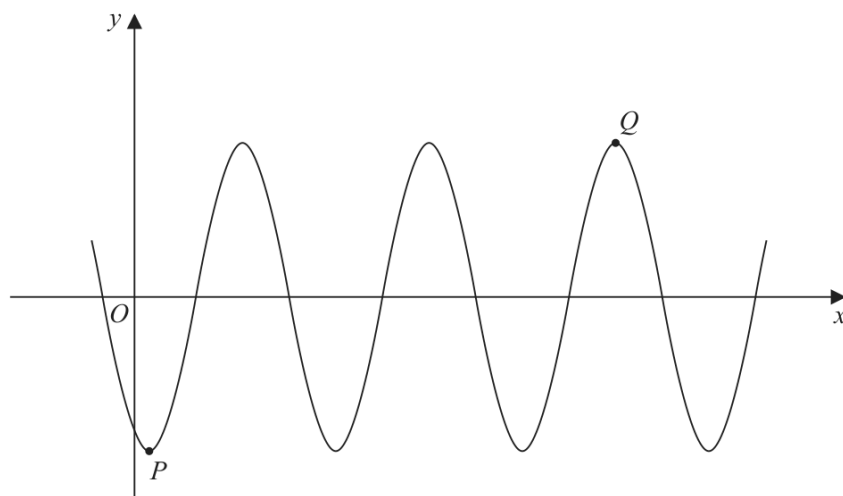
**Figure 4**

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A .

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)**(Total 4 marks)**

9. (i)

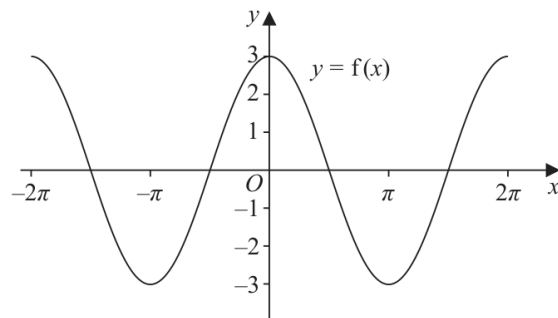
**Figure 3**

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

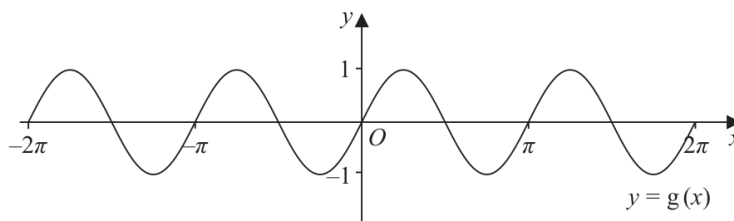
**Figure 4**

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(3)**(Total for Question 9 is 9 marks)**

10.

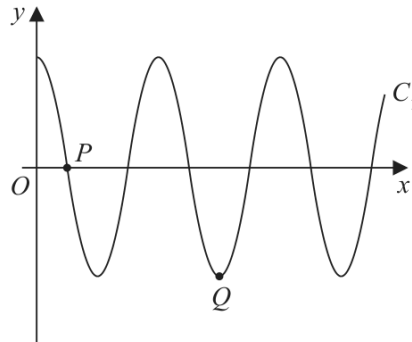
**Figure 4**

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

(Total for Question 10 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q6

P1 · TOPIC 11 6 marks

6.

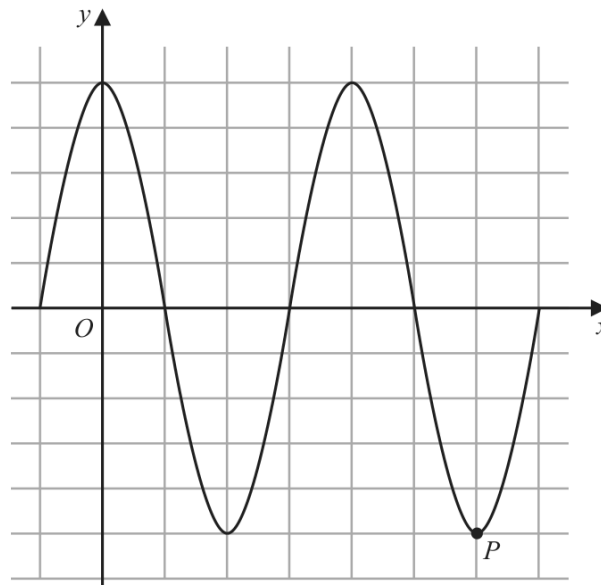
**Figure 2**

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

(Total for Question 6 is 6 marks)

Q20 PAST PAPER Pearson IAL WMA11 · June 2024 · Q11

P1 · TOPIC 11 6 marks

11.

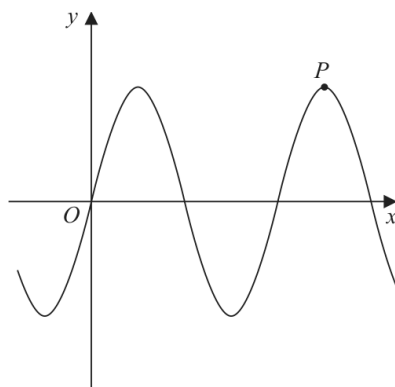


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1 .

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

4.

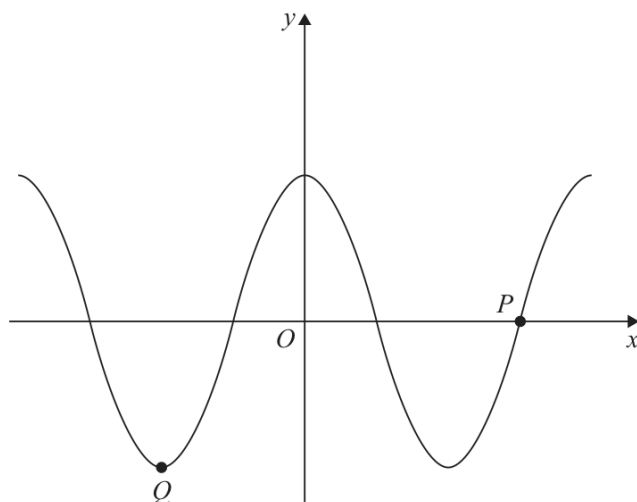


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = 4 \cos x$$

where x is measured in degrees.

The points P and Q lie on the curve and are shown in Figure 1.

(a) State the coordinates of P .

(1)

(b) State the coordinates of Q .

(2)

(c) State the **number** of solutions of the equation

(i) $4 \cos x = 3$ in the interval $0 < x \leq 18\,000^\circ$

(ii) $5 + 4 \cos x = 1$ in the interval $-720^\circ < x \leq 720^\circ$

(iii) $4 \cos x - 3 = 1$ in the interval $-1080^\circ \leq x \leq 1080^\circ$

(3)

(Total for Question 4 is 6 marks)

03 Solving by Symmetry

Your calculator gives one answer. The graph gives all the others.

WHEN YOU SEE → YOU DO

“given that $\sin 60^\circ = \dots$, find all values ...”

→ use the symmetry of the graph

“in terms of α ” / “in terms of p ”

→ write the other solutions from α

$\sin 3x = \dots$ for $0 \leq x \leq 2\pi$

→ solve for $3x$ in $0 \leq 3x \leq 6\pi$, then divide

$$\sin x = k \quad \alpha \quad 180^\circ - \alpha$$

$$\cos x = k \quad \alpha \quad 360^\circ - \alpha \text{ (or } -\alpha)$$

$$\tan x = k \quad \alpha \quad \alpha + 180^\circ$$

Then add or subtract whole periods to stay in the interval.

WORKED MODEL • JUNE 2022

$\sin \alpha = p$. Find, in terms of α , the solutions of $\sin 2x = p$ for $0 < x < 180^\circ$.

1 $2x = \alpha$ or $2x = 180^\circ - \alpha$

symmetry of sin

2 range of $2x$: $0 < 2x < 360^\circ$

no more in range

ANSWER $x = \frac{\alpha}{2}$ or $x = 90^\circ - \frac{\alpha}{2}$

IDEA 03

Practise: Solving by Symmetry

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM Symmetry of sine

P1 • TOPIC 11 2 marks

Given $\sin 30^\circ = 0.5$, find all solutions of $\sin x = 0.5$ for $0 \leq x \leq 720^\circ$.

Q23 DR ESLAM Inside a bracket

P1 • TOPIC 11 3 marks

Solve $\cos 2x = 0$ for $0 \leq x \leq 360^\circ$.

Work with $2x$ in $0 \leq 2x \leq 720^\circ$ first.

SELECTED PRACTICE

Q24

PRACTICE

Practice · Easy

P1 · TOPIC 11 2 marks

- 5 Write down the values for which $\cos x = \frac{1}{2}$, for $0 \leq x \leq 2\pi$.

(2 marks)

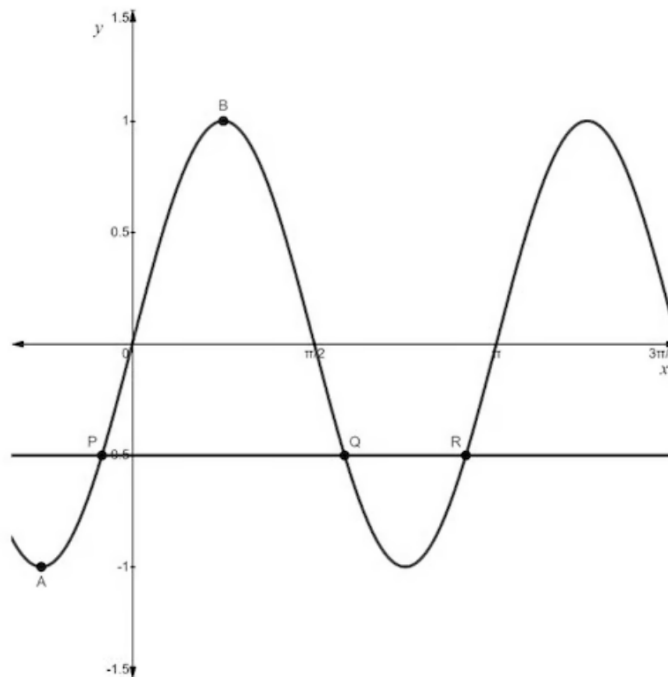
Q25**PRACTICE**

Practice • Medium

P1 • TOPIC 11 3 marks

8 (a) The graph below shows the curve with equation $y = \sin 2x$ in the interval

$$-\frac{\pi}{3} \leq x \leq \frac{3\pi}{2}.$$



Point A has coordinates $\left(-\frac{\pi}{4}, -1\right)$ and is the minimum point closest to the origin.

Point B is the maximum point closest to the origin. State the coordinates of B .

(1 mark)

(b) A straight line with equation $y = -\frac{1}{2}$ meets the graph of $y = \sin 2x$ at the three points P , Q and R , as shown in the diagram.

Given that point P has coordinates $\left(-\frac{\pi}{12}, -\frac{1}{2}\right)$, use graph symmetries to determine the coordinates of Q and R .

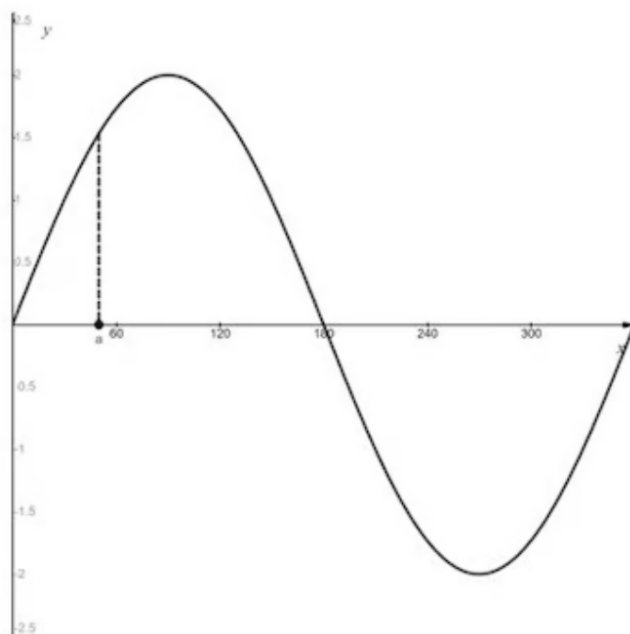
(2 marks)

Q26**PRACTICE**

Practice • Very Hard

P1 • TOPIC 11 2 marks

- 9 The graph below shows the curve with equation $y = 2 \sin \theta$, in the interval $0^\circ \leq \theta \leq 360^\circ$. One value of θ has been labelled ($\theta = a^\circ$).



Use the graph, along with the symmetry properties of the sine function, to verify that

$$2 \sin a = 2 \sin(180^\circ - a) = -2 \sin(180^\circ + a) = -2 \sin(360^\circ - a).$$

(2 marks)

PAST PAPERS

Q27 PAST PAPER Pearson IAL WMA11 · June 2022 · Q9

P1 · TOPIC 11 8 marks

9.

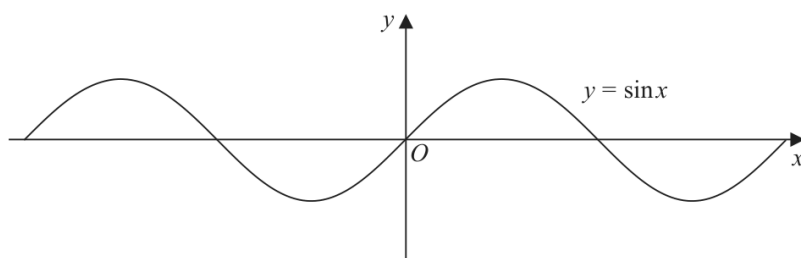


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

(a) state, in terms of p , the value of

(i) $2 \sin(180^\circ - \alpha)$

(ii) $\sin(\alpha - 180^\circ)$

(iii) $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

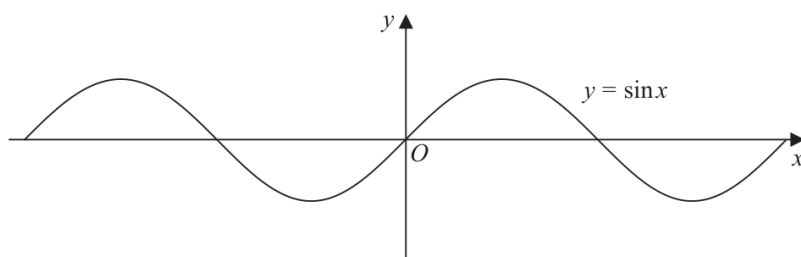


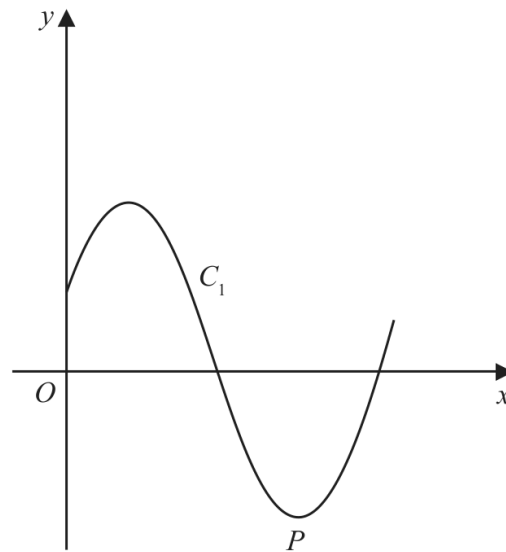
Diagram 1

(Total 8 marks)

Q28 PAST PAPER Pearson IAL WMA11 · June 2026 · Q9

P1 · TOPIC 11 5 marks

9.

**Figure 4**

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leq x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(a) State the x coordinate of P .

(1)

(b) State the number of roots of the equation

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leq x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leq x < 2\pi$

(c) State the coordinates of Q .

(2)**Question 9 continued****(Total for Question 9 is 5 marks)**

04 Counting Solutions: Two Graphs

“How many solutions” is never solved — it is **counted**. Draw both sides as graphs and count the crossings.

WHEN YOU SEE → YOU DO

“state the number of solutions/roots”

→ sketch both sides, count intersections

“giving a reason”

→ “the graphs intersect n times”

a huge interval ($-100\pi \leq x \leq 100\pi$)

→ count in one period, multiply, fix the ends

$\tan x =$ **bounded curve**

→ one crossing on every branch

1

Rearrange

Put the equation as *graph you have* = *graph you can draw*.

2

Count one period

Mark the crossings in one period.

3

Scale up, check the ends

Multiply by the number of periods; look carefully at the first and last part-period.

WORKED MODEL • JANUARY 2023

$$\tan x = \pi x \text{ in } -100\pi < x < 100\pi$$

1 central branch: $x = 0$ and one each side

3 roots

2 every other full branch: the line crosses once

99 each side

ANSWER $3 + 2 \times 99 = 201$

IDEA 04

Practise: Counting Solutions

My questions first, then selected practice, then real past papers.

Q29**DR ESLAM**

Count, do not solve

P1 • TOPIC 11 2 marks

By sketching $y = \sin x$ and $y = \frac{x}{360}$ for $0 \leq x \leq 360^\circ$, state the number of solutions of $\sin x = \frac{x}{360}$.

SELECTED PRACTICE**Q30****PRACTICE**

Practice • Medium

P1 • TOPIC 11 5 marks

- 6 (i) On the same set of axes, sketch the graphs of $y = \cos \theta$ and $y = \cos 3\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$, giving the coordinates of all points of intersection with the coordinate axes.
- (ii) State the coordinates of any points where the graphs intersect.

(5 marks)

Q31**PRACTICE**

Practice · Hard

P1 · TOPIC 11 8 marks

- 12 (i) On the same set of axes, sketch the graphs of $y = \tan \frac{1}{4} \theta$ and $y = \cos(\theta + 120^\circ)$ in the interval $0^\circ \leq \theta \leq 270^\circ$. Show clearly the coordinates of all points of intersection with the coordinate axes.
- (ii) Deduce the number of solutions to the equation $\cos(\theta + 120^\circ) - \tan \frac{1}{4} \theta = 0$, in the interval $0^\circ \leq \theta \leq 270^\circ$.

(8 marks)

Q32**PRACTICE**

Practice · Very Hard

P1 · TOPIC 11 7 marks

- 4 (a) On the same set of axes, sketch the graphs of $y = \tan \frac{1}{2}\theta$ and $y = \tan\left(\theta - \frac{\pi}{6}\right)$ in the interval $-2\pi \leq \theta \leq 2\pi$. Label the coordinates of points of intersection with the coordinate axes.

(4 marks)

- (b) Within the interval $-2\pi \leq \theta \leq 2\pi$, determine the coordinates of the two points where $\tan \frac{1}{2}\theta = \tan\left(\theta - \frac{\pi}{6}\right)$. Give your answer in surd form.

(3 marks)

PAST PAPERS

Q33 PAST PAPER Pearson IAL WMA11 • June 2021 • Q9

P1 • TOPIC 11 7 marks

9.

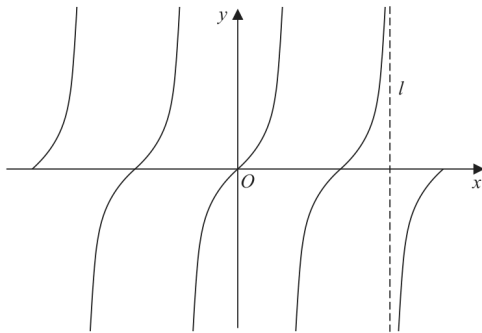


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

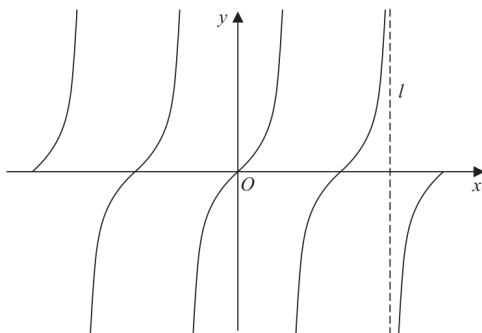


Diagram 1

(Total 7 marks)

Q34 PAST PAPER Pearson IAL WMA11 · January 2023 · Q9

P1 · TOPIC 11 4 marks

9.

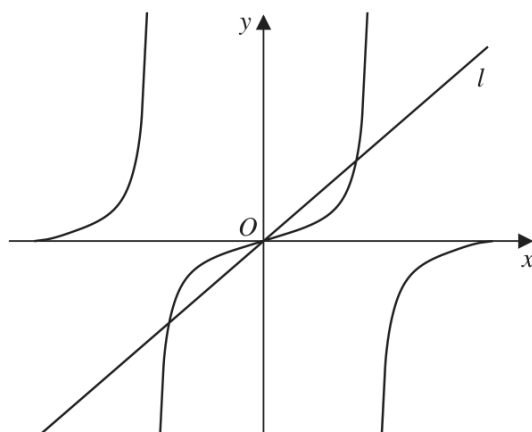


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

Q35 PAST PAPER Pearson IAL WMA11 · October 2024 · Q7

P1 · TOPIC 11 8 marks

7.

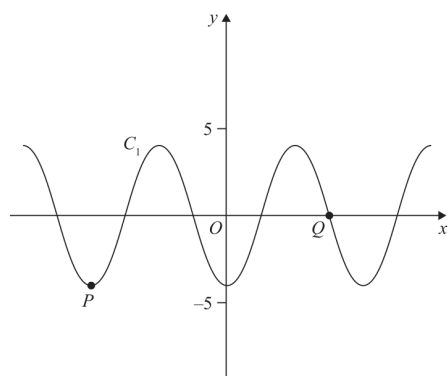
**Figure 3**

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4 \cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

(i) the coordinates of P

(ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4 \cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

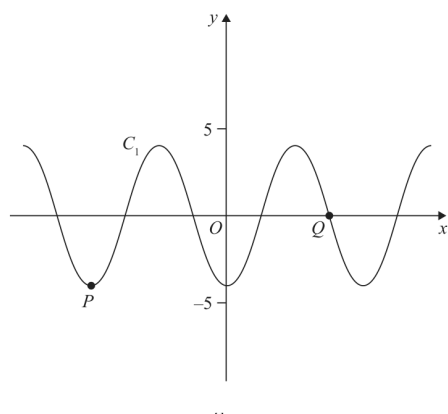
(ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi}x$$



(Total for Question 7 is 8 marks)

Q36**PAST PAPER**

Pearson IAL WMA11 • January 2026 • Q9

P1 • TOPIC 11 9 marks

9.

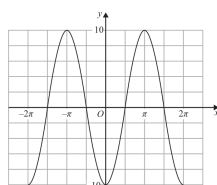
**Figure 4**

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(ii) Hence find the **number** of solutions of the equation

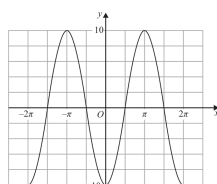
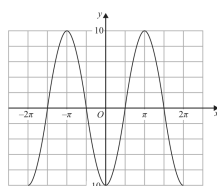
$$f(x) = \tan x \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

giving a reason for your answer.

(4)

(a)

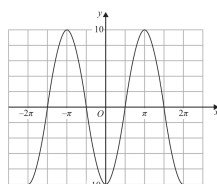
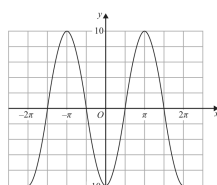
(b)(i)

**Diagram 1****Copy of Diagram 1**

Only use this copy if you need to redraw your graph.

(b)(ii)

(c)(i)

**Diagram 2****Copy of Diagram 2**

Only use this copy if you need to redraw your graph.

(c)(ii)

(Total for Question 9 is 9 marks)

05 Modelling with Trig

Tides, rollercoasters, dips in the ground. Same graph — the question is about its period and its max/min.

WHEN YOU SEE → YOU DO

“ $y = 30 \cos(24t)$ ”

→ period = $\frac{360}{24} = 15$ s

“how many dips / falls”

→ count minima in the interval

“at or above halfway”

→ solve $\sin(\dots) \geq \frac{1}{2}$ with the graph

WORKED MODEL • TIDAL ESTUARY

$y = \sin\left(\frac{\pi}{8}t\right)$; when is $y \geq \frac{1}{2}$?

1 $\frac{\pi}{8}t = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

symmetry

2 $t = \frac{4}{3}$ to $t = \frac{20}{3}$ hours

solve

ANSWER about 3:30 pm to 8:30 pm

IDEA 05

Practise: Modelling with Trig

My questions first, then selected practice.

Q37 DR ESLAM Period of a model

P1 • TOPIC 11 2 marks

The height of a wave is $h = 2 \sin(30t)$ metres, t in seconds. Find the period and the greatest height.

SELECTED PRACTICE

Q38

PRACTICE

Practice · Medium

P1 · TOPIC 11 6 marks

- 11** A section of a new rollercoaster has a series of rises and falls. The vertical displacement of the rollercoaster carriage, y , measured in metres relative to a fixed reference height, can be modelled using the function $y = 30 \cos(24t)^\circ$, where t is the time in seconds.
- (i) Sketch the function for the interval $0 \leq t \leq 30$.
- (ii) How many times will the rollercoaster carriage fall during the 30 seconds?
- (iii) How long does the model suggest it will take for the rollercoaster carriage to reach the bottom of the first fall?

(6 marks)

Q39**PRACTICE**

Practice · Hard

P1 · TOPIC 11 5 marks

- 10** Changes in the depth of water in a small tidal estuary relative to a fixed reference depth can be modelled using the function $y = \left(\sin \frac{\pi}{8} t \right)$, where y is measured in metres, the angle is measured in radians, and t is the time in hours.

(i) Sketch the function for the interval $0 \leq t \leq 8$.

(ii) If $t = 0$ represents 2pm, during what times, to the nearest half hour, will the estuary be at or above the halfway point between $y = 0$ and its maximum depth?

(5 marks)

Q40**PRACTICE**

Practice · Hard

P1 · TOPIC 11 5 marks

11 A series of dips and mounds caused by underground mining has a cross-section which can be modelled using the function $y = 4\cos(18x)^\circ$, where x and y are respectively the horizontal and vertical displacements, in metres, from a fixed origin point.

(i) Sketch the function for the interval $0 \leq x \leq 40$ and state the periodicity of the model.

(ii) How many dips are in this model in the given interval?

(5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|--|
| 01 | Three graphs | Five key points each. Period $360^\circ / 360^\circ / 180^\circ$. |
| 02 | $a \sin(bx + c) + k$ | a height $\cdot b$ squash (period = $\frac{360}{b}$) $\cdot c$ opposite shift $\cdot k$ up. |
| 03 | Symmetry | $\sin: 180 - \alpha \cdot \cos: 360 - \alpha \cdot \tan: \alpha + 180$. |
| 04 | Counting | Two graphs $\rightarrow$ count crossings $\rightarrow$ per period $\rightarrow$ the ends. |
| 05 | Models | Period and max/min answer the story. |

BOTTOM LINE Sketch first $\rightarrow$ move the key points $\rightarrow$ one answer from the calculator, the rest from symmetry $\rightarrow$ count, do not solve.

Answers

Trigonometric Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE THREE GRAPHS

- Q1** Dr Eslam • Five key points $(0, 1), (90, 0), (180, -1), (270, 0), (360, 1)$
- Q2** Dr Eslam • Periods and asymptotes $180^\circ; x = 90^\circ, x = 270^\circ$
- Q3** Practice • Easy
- (i) zeros $-180, 0, 180$; max $(90, 1)$, min $(-90, -1)$ (ii) $(0, 1), (90, 0), (180, -1), (270, 0), (360, 1)$ (iii) zeros $-180, 0, 180$; asymptotes $x = \pm 90$
- Q4** Practice • Medium (i) $(0, 1), (90, 0), (270, 0)$ (ii) $90^\circ, 270^\circ$
- Q5** Practice • Hard (ii) $-60^\circ, -120^\circ$

IDEA 02 • TRANSFORMATIONS

- Q6** Dr Eslam • Read a and k $k = 3$; minimum -1
- Q7** Dr Eslam • Move a point (a) $(90, 3)$ (b) $(45, 1)$ (c) $(120, 1)$
- Q8** Dr Eslam • Period from b (a) 120° (b) 720°
- Q9** Practice • Easy (i) $(90, 3)$ (ii) $(30, 1)$ (iii) $(60, 1)$
- Q10** Practice • Hard (ii) $-\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$
- Q11** Practice • Very Hard (i) $p = 5$ (ii) period $\frac{2\pi}{5}$
- Q12** Pearson IAL WMA11 • October 2019 • Q9 (a) 24π (b) $(18\pi, -1)$ (c)(i) $-12\pi - \alpha$ (ii) $6\pi - \alpha$
- Q13** Pearson IAL WMA11 • January 2020 • Q7 (a) $P(90, 3), Q(360, 0)$ (b) $R(270, 4)$
- Q14** Pearson IAL WMA11 • January 2021 • Q3 (a) $P(-180, -4), Q(450, 0)$ (b) $R(360, 7)$
- Q15** Pearson IAL WMA11 • October 2021 • Q4 (a) $Q(90, -1), k = 225$ (b) $-1 < p < 0$ or $p = 1$
- Q16** Pearson IAL WMA11 • January 2022 • Q9 (a) $A = -3$ (b) $Q(930, 3)$
- Q17** Pearson IAL WMA11 • June 2023 • Q9 (i) $f(x) = 3 \cos x$; y -int $\frac{3\sqrt{2}}{2}$ (ii) $g(x) = \sin 2x$; y -int -2
- Q18** Pearson IAL WMA11 • October 2023 • Q10 (a) $n = 3$, period 1080° (b) $(540, -3)$ (c) $k = 0.9$
- Q19** Pearson IAL WMA11 • January 2024 • Q6 (a) $(540, -5)$ (b)(i) $(360, 3)$ (ii) $(180, 5)$
- Q20** Pearson IAL WMA11 • June 2024 • Q11 (a) $(\frac{5\pi}{2}, 12)$ (b) $(\frac{3\pi}{2}, -21)$ (c) $A = -12, B = \frac{5\pi}{4}$
- Q21** Pearson IAL WMA11 • June 2025 • Q4 (a) $(270, 0)$ (b) $(-180, -4)$ (c) $100, 4, 7$

IDEA 03 • SOLVING BY SYMMETRY

- Q22** Dr Eslam • Symmetry of sine 30, 150, 390, 510
- Q23** Dr Eslam • Inside a bracket 45, 135, 225, 315
- Q24** Practice • Easy $\frac{\pi}{3}, \frac{5\pi}{3}$
- Q25** Practice • Medium (a) $B(\frac{\pi}{4}, 1)$ (b) $Q(\frac{7\pi}{12}, -\frac{1}{2}), R(\frac{11\pi}{12}, -\frac{1}{2})$
- Q26** Practice • Very Hard verified by the symmetry of sin
- Q27** Pearson IAL WMA11 • June 2022 • Q9 (a) $2p, -p, 3 - p$ (c) $\frac{a}{2}, 90^\circ - \frac{a}{2}$
- Q28** Pearson IAL WMA11 • June 2026 • Q9 (a) $\frac{11\pi}{8}$ (b) 102 (c) $Q(\frac{3\pi}{8}, -5)$

IDEA 04 • COUNTING SOLUTIONS

- Q29** Dr Eslam • Count, do not solve 2 (at $x = 0$ and once in $(90, 180)$)
- Q30** Practice • Medium (ii) $(-180, -1), (-90, 0), (0, 1), (90, 0), (180, -1)$
- Q31** Practice • Hard (ii) no solutions
- Q32** Practice • Very Hard (b) $(\frac{\pi}{3}, \frac{\sqrt{3}}{3}), (-\frac{5\pi}{3}, \frac{\sqrt{3}}{3})$
- Q33** Pearson IAL WMA11 • June 2021 • Q9 (a) $x = \frac{3\pi}{2}$ (b) asymptote $y = 1$; 5 solutions
(c)(i) 40 (ii) 14
- Q34** Pearson IAL WMA11 • January 2023 • Q9 (a) π (b) 3, 5, 201
- Q35** Pearson IAL WMA11 • October 2024 • Q7 (a) $P(-2\pi, -4), Q(\frac{3\pi}{2}, 0)$ (b) $k = 7$; $(2\pi, 3)$ (c) 1 root
- Q36** Pearson IAL WMA11 • January 2026 • Q9 (a) $f(x) = -10 \cos x$ (b)(ii) 2 (c)(ii) 200

IDEA 05 • MODELLING WITH TRIG

- Q37** Dr Eslam • Period of a model period 12 s; greatest height 2 m
- Q38** Practice • Medium (i) period 15 s, y from 30 to -30 (ii) 2 (iii) 7.5 s
- Q39** Practice • Hard (ii) about 3:30 pm to 8:30 pm
- Q40** Practice • Hard (i) period 20 m (ii) 2 dips